

Whitehead Torsion and the s-Cobordism Theorem

1 Framework

Let $(W; M_0, M_1)$ be a smooth h-cobordism: W is a compact connected smooth manifold with $\partial W = M_0 \sqcup M_1$, and both boundary inclusions are homotopy equivalences. Throughout, the boundary manifolds are closed and connected, and

$$n = \dim W = \dim M_0 + 1 \geq 6, \quad \pi = \pi_1(M_0) \cong \pi_1(W).$$

No orientability assumption is needed. A diffeomorphism relative to M_0 restricts to $x \mapsto (x, 0)$ on the specified incoming boundary; the outgoing boundary has no additional fixed parametrization.

The smooth s-cobordism theorem identifies Whitehead torsion as the obstruction to a product structure:

$$W \cong M_0 \times [0, 1] \text{ rel } M_0 \iff \tau(W, M_0) = 0 \in \text{Wh}(\pi).$$

Its classification form states that h-cobordisms over M_0 , up to diffeomorphism relative to M_0 , are classified by $\text{Wh}(\pi)$.

We follow the handle-theoretic treatment in [1, Chapters 2–3]; a classical account is [4]. The geometric normal form, modification, homology, and cancellation lemmas are taken as inputs from the preceding part of the proof. The argument below identifies the remaining handle matrix with Whitehead torsion and proves the vanishing, realization, and classification statements. Supporting algebraic and simple-homotopy arguments are collected in the appendices.

1.1 Handle normal form

Handle trading and cancellation, including the high-dimensional Whitney trick, give the following normal form [1, Chapter 2].

Lemma 1.1 (Normal form lemma). *Let $(W; M_0, M_1)$ be an n -dimensional smooth h-cobordism with $n \geq 6$. For any integer q satisfying*

$$2 \leq q \leq n - 3,$$

there is a handle decomposition relative to M_0 of the form

$$W \cong M_0 \times [0, 1] + \sum_{i=1}^r (\phi_i^q) + \sum_{j=1}^s (\phi_j^{q+1}).$$

That is, all handles have index q or $q + 1$.

The notation $+\sum(\phi_i^q)$ means that q -handles $D^q \times D^{n-q}$ are attached by framed embeddings $\phi_i^q : S^{q-1} \times D^{n-q} \hookrightarrow \partial_1 W_{q-1}$. Here $\partial_1 W_{q-1}$ is the outgoing boundary after the preceding handles have been attached; corners are smoothed after attachment.

The word “middle” refers to these consecutive indices, which need not be the exact middle of n . Since the relative handle complex is acyclic, its sole differential is an isomorphism. Applying the augmentation $\mathbb{Z}\pi \rightarrow \mathbb{Z}$ shows that $r = s$. Cancelling these remaining pairs would leave a product collar, so the problem is to determine the obstruction to their cancellation.

2 The Whitehead Group

Before explaining the obstruction, we recall the definition of the Whitehead group.

2.1 $K_1(R)$

Let R be an associative ring with unit. For every $n \geq 1$, let

$$\mathrm{GL}_n(R)$$

be the group of invertible $n \times n$ matrices over R .

There are stabilization maps

$$\mathrm{GL}_n(R) \longrightarrow \mathrm{GL}_{n+1}(R), \quad A \longmapsto \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

Define

$$\mathrm{GL}(R) = \varinjlim_n \mathrm{GL}_n(R).$$

Then

$$K_1(R) := \mathrm{GL}(R)_{\mathrm{ab}} = \mathrm{GL}(R) / [\mathrm{GL}(R), \mathrm{GL}(R)].$$

We write $K_1(R)$ additively, so $[AB] = [A] + [B]$ and $[A^{-1}] = -[A]$. The stable Whitehead lemma identifies the commutator subgroup with the subgroup $E(R)$ generated by elementary matrices $I + aE_{ij}$, where $i \neq j$ and $a \in R$ [1, Chapter 3]. Thus $K_1(R)$ records invertible matrices up to stabilization and elementary operations. This algebraic fact is needed when we later turn vanishing in $\mathrm{Wh}(\pi)$ into a sequence of handle operations.

2.2 Trivial units

Now let G be a group and take

$$R = \mathbb{Z}G.$$

The group ring $\mathbb{Z}G$ consists of finite sums $\sum_g n_g g$, with $n_g \in \mathbb{Z}$, and multiplication extends the group multiplication distributively. Every element

$$\pm g, \quad g \in G,$$

is a unit in the group ring $\mathbb{Z}G$. Hence it defines an invertible 1×1 matrix

$$(\pm g).$$

Therefore we get a homomorphism

$$G \times \{\pm 1\} \longrightarrow K_1(\mathbb{Z}G), \quad (g, \pm 1) \longmapsto [(\pm g)].$$

Definition 2.1 (Whitehead group). The Whitehead group of G is

$$\mathrm{Wh}(G) := \mathrm{coker}(G \times \{\pm 1\} \longrightarrow K_1(\mathbb{Z}G)).$$

Equivalently,

$$\mathrm{Wh}(G) = K_1(\mathbb{Z}G) / \langle [(\pm g)] \mid g \in G \rangle.$$

Thus $\mathrm{Wh}(G)$ is obtained from $K_1(\mathbb{Z}G)$ by killing the obvious units coming from signs and group elements.

2.3 Matrix interpretation

For the proof of the s-cobordism theorem, it is useful to think of an element of $\text{Wh}(G)$ as represented by an invertible matrix

$$A \in \text{GL}_r(\mathbb{Z}G),$$

where two such matrices represent the same class if a finite sequence of the following operations and their inverses connects them. This description uses the stable Whitehead lemma:

(i) stabilization:

$$A \sim \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix};$$

(ii) elementary row or column additions: add a left multiple of one row to another row, or a right multiple of one column to another column;

(iii) permutation of rows or columns;

(iv) multiplying a row on the left, or a column on the right, by a trivial unit

$$\pm g, \quad g \in G.$$

The order of multiplication matters when G is nonabelian. The identity matrix, including the empty 0×0 matrix, represents zero. Addition in $\text{Wh}(G)$ is represented by block sum, and $[A^{-1}] = -[A]$. Thus $\text{Wh}(G)$ records the class that remains after these specified changes of basis; an arbitrary change of basis need not have zero Whitehead class.

3 The Handle Matrix

Consider the h-cobordism

$$(W; M_0, M_1).$$

Suppose that W has been put into normal form:

$$W \cong M_0 \times [0, 1] + \sum_{i=1}^r (\phi_i^q) + \sum_{j=1}^r (\phi_j^{q+1}).$$

Let

$$\widetilde{W} \longrightarrow W$$

be the universal cover. Since

$$\pi_1(W) \cong \pi,$$

the deck transformation group is π . We let it act on the left. The inverse image $\widetilde{M}_0 \subset \widetilde{W}$ is connected and is the universal cover of M_0 , because the boundary inclusion induces an isomorphism on fundamental groups.

A handle decomposition determines a relative CW model, with one cell for each handle. In this report

$$C_*(\widetilde{W}, \widetilde{M}_0)$$

denotes the relative *handle* chain complex, or equivalently the cellular complex of that model, rather than the cellular complex of an arbitrary triangulation of W . The comparison with the manifold's simple homotopy type is explained in Appendix B. Thus the relative chain complex

$$C_*(\widetilde{W}, \widetilde{M}_0)$$

is a chain complex of free left $\mathbb{Z}\pi$ -modules.

Choose an orientation and one lift of every handle. These choices give bases:

- each q -handle gives a basis vector of $C_q(\widetilde{W}, \widetilde{M}_0)$;
- each $(q+1)$ -handle gives a basis vector of $C_{q+1}(\widetilde{W}, \widetilde{M}_0)$.

Because there are no other handles, the chain complex has the form

$$0 \longrightarrow C_{q+1}(\widetilde{W}, \widetilde{M}_0) \xrightarrow{d_{q+1}} C_q(\widetilde{W}, \widetilde{M}_0) \longrightarrow 0.$$

Since $M_0 \hookrightarrow W$ is a homotopy equivalence, its lift $\widetilde{M}_0 \hookrightarrow \widetilde{W}$ is also a homotopy equivalence. In particular,

$$H_*(\widetilde{W}, \widetilde{M}_0) = 0.$$

Therefore the above two-term chain complex is acyclic.

It follows that

$$d_{q+1} : C_{q+1}(\widetilde{W}, \widetilde{M}_0) \longrightarrow C_q(\widetilde{W}, \widetilde{M}_0)$$

is an isomorphism.

Choosing the handle bases, we may identify

$$C_{q+1}(\widetilde{W}, \widetilde{M}_0) \cong (\mathbb{Z}\pi)^r, \quad C_q(\widetilde{W}, \widetilde{M}_0) \cong (\mathbb{Z}\pi)^r.$$

Then the differential d_{q+1} is represented by an invertible matrix

$$A \in \mathrm{GL}_r(\mathbb{Z}\pi).$$

We use row coordinates for based left modules: if $d(e_j) = \sum_i a_{ji}e_i$, then d takes a row of coefficients x to xA . This is left $\mathbb{Z}\pi$ -linear even when the ring is noncommutative. We keep this convention for all matrices below.

This matrix records the differential between the remaining handles.

3.1 Intersection coefficients

Let

$$[\phi_i^q] \in C_q(\widetilde{W}, \widetilde{M}_0)$$

be the basis element corresponding to the i -th q -handle, and let

$$[\phi_j^{q+1}] \in C_{q+1}(\widetilde{W}, \widetilde{M}_0)$$

be the basis element corresponding to the j -th $(q+1)$ -handle.

The differential has the form

$$d_{q+1}[\phi_j^{q+1}] = \sum_{i=1}^r a_{ji}[\phi_i^q], \quad a_{ji} \in \mathbb{Z}\pi.$$

Thus the matrix

$$A = (a_{ji})$$

records how the attaching spheres of the $(q+1)$ -handles algebraically intersect the transverse spheres of the q -handles.

More precisely, let N be the outgoing level boundary after the q -handles have been attached. The attaching sphere α_j of a $(q+1)$ -handle has dimension q , and the belt sphere (also called transverse sphere) β_i of a q -handle has dimension $n - q - 1$. Hence

$$\dim \alpha_j + \dim \beta_i = q + (n - q - 1) = \dim N.$$

After a transverse perturbation their intersections are isolated. With paths to a common basepoint, orientations, and lifts chosen consistently, the coefficient is

$$a_{ji} = \sum_{p \in \alpha_j \cap \beta_i} \epsilon(p) g(p) \in \mathbb{Z}\pi,$$

where $\epsilon(p) \in \{\pm 1\}$ is the local intersection sign and $g(p)$ records the corresponding deck transformation. For a nonorientable manifold, signs are defined using the lifted handle orientations and transport along the chosen paths. The coefficient is the cellular boundary coefficient, so no global orientation is required [1, Chapter 2].

Thus:

- a q -handle gives a transverse sphere;
- a $(q + 1)$ -handle has an attaching sphere;
- the coefficient $a_{ji} \in \mathbb{Z}\pi$ records the signed intersections between the attaching sphere of the j -th $(q + 1)$ -handle and the transverse sphere of the i -th q -handle, together with fundamental group information.

If the matrix were the identity matrix

$$A = I_r,$$

then $a_{ji} = \delta_{ji}$ in the group ring. This is stronger than an intersection count after augmentation: intersections cancel separately for each group element. The homology lemma and the Whitney trick in the normal form range $n \geq 6$, $2 \leq q \leq n - 3$ convert this algebraic condition into the following geometric one [1, Chapter 2]: the attaching sphere can be isotoped so that it meets the corresponding transverse sphere in exactly one point and misses all other transverse spheres.

The relevant Whitney moves remove opposite-sign intersections with matching group labels; the geometric homology lemma supplies the required embedded, framed Whitney disks and the relative isotopies. An algebraic intersection number of one alone is not the cancellation hypothesis. After this geometric adjustment, the cancellation lemma applies and the two handles cancel [1, Chapter 2].

So the identity matrix corresponds geometrically to perfect handle cancellation.

3.2 Handle operations

The matrix A itself depends on choices:

- ordering of the handles;
- choices of lifts to the universal cover;
- choices of orientations;
- changes of handle decomposition by handle slides and adding cancelling pairs.

For a fixed value of q , handle slides, changes of lifts and orientations, and addition of cancelling $(q, q + 1)$ -pairs modify A by the operations listed in Section 2. Thus the class

$$[A] \in \text{Wh}(\pi)$$

is unchanged under these modifications. Independence of an arbitrary choice of handle decomposition will follow from its identification with $(-1)^q \tau(W, M_0)$ in Section 4.5. In particular, the vanishing of this class is independent of the index chosen for normal form.

The geometric meaning of vanishing is established in Section 5.

4 Whitehead Torsion

4.1 Contractible complexes

For the torsion discussion, let R be an associative ring with unit and invariant basis number: $R^a \cong R^b$ implies $a = b$. This ensures that an isomorphism between based free modules has a square matrix. In our application,

$$R = \mathbb{Z}\pi, \quad \pi = \pi_1(M_0).$$

The group ring has invariant basis number, as one sees by applying its augmentation to $\mathbb{Z}$. A finite based free R -chain complex is a bounded chain complex

$$C_* = (\cdots \rightarrow C_{p+1} \xrightarrow{c_{p+1}} C_p \xrightarrow{c_p} C_{p-1} \rightarrow \cdots)$$

such that each C_p is a finitely generated free R -module together with a chosen ordered basis. Here “bounded” means that only finitely many degrees have nonzero modules. Acyclic means that all homology groups are zero.

Suppose that C_* is contractible. This means that there exists a chain contraction, namely maps

$$\gamma_p : C_p \longrightarrow C_{p+1}$$

such that

$$c_{p+1}\gamma_p + \gamma_{p-1}c_p = \text{id}_{C_p}$$

for all p .

Define

$$C_{\text{odd}} = \bigoplus_p C_{2p+1}, \quad C_{\text{ev}} = \bigoplus_p C_{2p}.$$

The differential c_* and the contraction γ_* combine to give an isomorphism

$$(c_* + \gamma_*)_{\text{odd}} : C_{\text{odd}} \longrightarrow C_{\text{ev}}.$$

Use the bases concatenated in increasing degree on each parity summand. The isomorphism is represented by an invertible matrix over R . Its class in $K_1(R)$ is independent of the chain contraction; this is the contraction-independence theorem for algebraic torsion [1, Chapter 3][3, Algebra]. Permuting the chosen bases can contribute a sign, so when no ordering convention is specified one works in

$$\overline{K}_1(R) = K_1(R) / \langle [(-1)] \rangle.$$

We also write $\Delta(C_*)$ for this reduced class when R is a general ring; for a group ring we pass to the further quotient below.

Definition 4.1 (Torsion convention). For $R = \mathbb{Z}\pi$, the torsion of a finite based free contractible complex is

$$\Delta(C_*) = [(c + \gamma)_{\text{odd}}] \in \text{Wh}(\pi).$$

Thus the convention is *odd to even*. Signs, reorderings, and changes of a basis vector by $\pm g$ disappear in the Whitehead group.

This is the algebraic torsion underlying Whitehead torsion. Reidemeister torsion often denotes a further specialization to a field or representation; we will keep the group-ring class itself.

Lemma 4.2. *A bounded acyclic chain complex of projective R -modules is contractible. In particular, this holds for a finite based free complex.*

Proof. Write $B_p = \text{im } c_{p+1} = \ker c_p$. Starting at the lowest nonzero degree, exactness gives split sequences

$$0 \longrightarrow B_p \longrightarrow C_p \xrightarrow{c_p} B_{p-1} \longrightarrow 0.$$

Indeed, the first nonzero B_p is the lowest module of the complex and is projective; induction makes each subsequent B_p a direct summand of C_p , hence projective. Choose splittings $C_p \cong B_p \oplus B_{p-1}$ for which $c_p(b_p, b_{p-1}) = (b_{p-1}, 0)$. Define

$$\gamma_p(b_p, b_{p-1}) = (0, b_p) \in B_{p+1} \oplus B_p = C_{p+1}.$$

Then $c\gamma + \gamma c = \text{id}$. □

This lemma justifies the use of “acyclic” in the handle calculation. It does not apply to arbitrary unbounded complexes or to arbitrary modules.

4.2 Mapping cones

Now let

$$f_* : C_* \longrightarrow D_*$$

be a chain homotopy equivalence of finite based free R -chain complexes. This means that a chain map g_* exists with $g_*f_* \simeq \text{id}_C$ and $f_*g_* \simeq \text{id}_D$, where chain homotopies raise degree by one. For a general R , the torsion below lies in $\overline{K}_1(R)$; for $R = \mathbb{Z}\pi$ we take its image in $\text{Wh}(\pi)$.

With D first in each degree, define the mapping cone by

$$\text{cone}(f_*)_p = D_p \oplus C_{p-1}, \quad \partial(y, x) = (d_D y + f_{p-1}(x), -d_C x).$$

The identity $\partial^2 = 0$ follows from the chain-map identity $d_D f = f d_C$. This cone is contractible; the cone criterion is proved in Appendix A. Therefore we define the Whitehead torsion of f_* by

$$\tau(f_*) := \Delta(\text{cone}(f_*)).$$

Thus Whitehead torsion measures the failure of a chain homotopy equivalence to be simple at the level of based free chain complexes.

4.3 Homotopy equivalences

Let

$$f : X \longrightarrow Y$$

be a homotopy equivalence of finite connected CW-complexes. Use a cellular approximation to f if necessary, and choose one oriented lift of each cell. A homotopy inverse and the lifted homotopies make the resulting cellular chain map a chain homotopy equivalence. Choose universal covers

$$\tilde{X} \longrightarrow X, \quad \tilde{Y} \longrightarrow Y.$$

After identifying fundamental groups using f , the cellular chain map

$$C_*(\tilde{f}) : C_*(\tilde{X}) \longrightarrow C_*(\tilde{Y})$$

is a chain homotopy equivalence of finite based free $\mathbb{Z}\pi_1(Y)$ -chain complexes. Therefore we define

$$\tau(f) := \tau(C_*(\tilde{f})) \in \text{Wh}(\pi_1(Y)).$$

The target group is used for the coefficients: more formally, the source complex is extended along the group-ring isomorphism induced by f . The class is independent of the cellular

approximation and is invariant under homotopy [1, Chapter 3][2]. Appendix C checks the choices of covers, lifts, and basepoints.

A homotopy equivalence is called *simple* if it is represented, up to homotopy, by a finite sequence of elementary CW expansions and collapses. The Whitehead torsion criterion says that this is equivalent to $\tau(f) = 0$ [3, Whitehead Torsion in the CW Category]. This criterion explains the term “s-cobordism”: it is an h-cobordism for which the incoming inclusion is simple.

For a nonconnected space, one defines the torsion componentwise. All spaces considered here are connected.

4.4 Cobordism torsion

Let

$$(W; M_0, M_1)$$

be an h-cobordism as in the introduction. Compact smooth manifolds admit finite CW structures coming from smooth triangulations. These give the canonical simple homotopy types used here; changing a compatible smooth triangulation preserves that type [2, 3]. Since

$$M_0 \hookrightarrow W$$

is a homotopy equivalence, its Whitehead torsion is defined.

Definition 4.3. The Whitehead torsion of the h-cobordism $(W; M_0, M_1)$ relative to M_0 is

$$\tau(W, M_0) := \tau(M_0 \hookrightarrow W) \in \text{Wh}(\pi_1 W).$$

Using the isomorphism

$$\pi_1(M_0) \cong \pi_1(W),$$

we regard it as an element

$$\tau(W, M_0) \in \text{Wh}(\pi_1 M_0).$$

Equivalently, one may compute this torsion from the relative chain complex

$$C_*(\widetilde{W}, \widetilde{M}_0).$$

For a CW pair, the mapping-cone torsion of an inclusion equals the torsion of its contractible relative complex. This follows from the additivity theorem and the zero torsion of the cone of the identity [1, Chapter 3]. The handle model is related to the manifold by a simple homotopy equivalence, so the same calculation applies to it. Appendix B supplies this comparison in detail.

For the proof of the s-cobordism theorem, this relative viewpoint is the most useful one, because the chain complex

$$C_*(\widetilde{W}, \widetilde{M}_0)$$

is exactly the handle chain complex associated to a handle decomposition of W relative to M_0 .

4.5 The torsion formula

In normal form the only differential is an isomorphism $d_{q+1} : C_{q+1} \rightarrow C_q$, represented by $A \in \text{GL}_r(\mathbb{Z}\pi)$. Its inverse gives a contraction. The relative-complex description therefore reduces cobordism torsion to the following parity calculation.

Proposition 4.4. *With the odd-to-even convention of Definition 4.1, a normal form with indices q and $q + 1$ satisfies*

$$\tau(W, M_0) = (-1)^q [A] \in \text{Wh}(\pi).$$

Proof. The only differential is $A : C_{q+1} \rightarrow C_q$, and a contraction is $A^{-1} : C_q \rightarrow C_{q+1}$. If q is even, the odd-to-even map is A . If q is odd, it is A^{-1} , whose class is $-[A]$. The preceding relative-complex identification gives the formula. $\square$

In particular, $q = 2$ is permitted whenever $n \geq 6$. The resulting 2/3 handle complex has torsion $[A]$. For any permitted q ,

$$\tau(W, M_0) = 0 \iff [A] = 0.$$

The sign $(-1)^q$ denotes addition or negation in the abelian group $\text{Wh}(\pi)$ and does not affect vanishing.

5 The s-Cobordism Theorem

Theorem 5.1 (s-Cobordism theorem). *Let $(W; M_0, M_1)$ be a smooth h -cobordism with $\dim M_0 \geq 5$. Then*

$$W \cong M_0 \times [0, 1] \text{ rel } M_0 \iff \tau(W, M_0) = 0 \in \text{Wh}(\pi_1 M_0).$$

Proof using the geometric lemmas. Suppose $\tau(W, M_0) = 0$. Choose a normal form with 2- and 3-handles. Its relative chain complex is

$$0 \longrightarrow C_3(\widetilde{W}, \widetilde{M}_0) \xrightarrow{A} C_2(\widetilde{W}, \widetilde{M}_0) \longrightarrow 0,$$

and Proposition 4.4 gives $[A] = 0$ in $\text{Wh}(\pi)$. The matrix description of the Whitehead group reduces A , after stabilization, to an identity matrix using elementary operations, permutations, and multiplication by trivial units.

The modification lemma realizes each elementary addition by finitely many handle slides, including slides along paths representing group elements. The other operations have the following geometric interpretations.

Elementary row operations correspond to sliding 3-handles over other 3-handles. Elementary column operations correspond to sliding 2-handles over other 2-handles. Stabilization corresponds to introducing a cancelling pair of handles. Multiplication by $\pm g$ corresponds to changing orientations and lifts of handles. Permutations correspond to renumbering handles.

Therefore we can change the handle decomposition, without changing the diffeomorphism type of W relative to M_0 , so that the new handle matrix is

$$I_{r'}.$$

Here r' is the number of pairs after any stabilizations. When the handle matrix is the identity, the attaching sphere of the j -th 3-handle algebraically intersects the transverse sphere of the j -th 2-handle exactly once, and algebraically misses all other transverse spheres.

In the level manifold of dimension $n - 1 \geq 5$, the attaching 2-spheres and belt $(n - 3)$ -spheres have complementary dimensions. The equality in $\mathbb{Z}\pi$ pairs the extra intersections with opposite signs and matching group labels. By the geometric homology lemma and the Whitney trick, this algebraic intersection statement can be improved to a geometric one: after isotopy, the attaching sphere of the j -th 3-handle meets the transverse sphere of the j -th 2-handle transversely in exactly one point and is disjoint from the transverse spheres of all other 2-handles.

Hence the cancellation lemma applies. The j -th 2-handle and the j -th 3-handle cancel.

Cancel all pairs successively. No handles remain, so the result is $M_0 \times [0, 1]$ relative to M_0 .

Conversely, suppose W is diffeomorphic to this product relative to M_0 . The incoming inclusion is identified with

$$M_0 \times \{0\} \hookrightarrow M_0 \times [0, 1].$$

A finite triangulated cylinder collapses to its incoming end after a compatible subdivision, so this is a simple homotopy equivalence [3]. A diffeomorphism preserves the canonical smooth simple homotopy type. Hence $\tau(W, M_0) = 0$. $\square$

6 Realization and Classification

6.1 Realization

Every Whitehead class occurs as the torsion of an h-cobordism [1, Theorem 2.1].

Theorem 6.1 (Realization). *Let M_0 be a closed connected smooth manifold with*

$$\dim M_0 \geq 5,$$

and put

$$\pi = \pi_1(M_0).$$

For every element

$$x \in \text{Wh}(\pi),$$

there exists a smooth h-cobordism

$$(W; M_0, M_1)$$

such that

$$\tau(W, M_0) = x.$$

Sketch of proof. Choose an invertible matrix

$$A \in \text{GL}_r(\mathbb{Z}\pi)$$

representing the class

$$x \in \text{Wh}(\pi).$$

The handle realization lemma supplies the geometric construction [1, Lemma 2.42(ii)]. Attach r trivial, pairwise disjoint, framed 2-handles to $M_0 \times \{1\}$. Their attaching loops are nullhomotopic, so this does not change the fundamental group. The lemma then realizes the rows of A by disjoint framed attaching 2-spheres for 3-handles. The existence of these embedded framed representatives is the geometric input; it does not follow from invertibility alone. This constructs a handle decomposition

$$W = M_0 \times [0, 1] + \sum_{i=1}^r (\phi_i^2) + \sum_{j=1}^r (\phi_j^3)$$

so that the associated relative handle chain complex is

$$0 \longrightarrow (\mathbb{Z}\pi)^r \xrightarrow{A} (\mathbb{Z}\pi)^r \longrightarrow 0.$$

Because A is invertible, this chain complex is contractible. The incoming inclusion induces a fundamental-group isomorphism and a homology isomorphism on universal covers. The homological Whitehead theorem therefore makes it a homotopy equivalence. From the other boundary, the dual handles have indices $n - 3$ and $n - 2$, both at least 3, so the outgoing inclusion also induces a fundamental-group isomorphism. The dual relative complex is acyclic by Poincaré–Lefschetz duality with local coefficients. Applying the same Whitehead theorem makes this inclusion a homotopy equivalence as well [1, Chapters 2–3]. Thus the result is an h-cobordism. Acyclicity alone, without these fundamental-group checks, would not suffice.

By construction, the Whitehead torsion is represented by A , and therefore

$$\tau(W, M_0) = [A] = x.$$

□

Thus every Whitehead class is realized by the remaining handle matrix of an h-cobordism. The next theorem also states that this class determines the h-cobordism relative to its incoming boundary.

6.2 Classification

We can now state the classification form [1, Theorem 2.1(iii)]. Two h-cobordisms over M_0 are equivalent if there is a diffeomorphism of their total spaces whose restriction to the specified incoming boundary is the identity. Their outgoing boundaries need not be specified in advance.

Theorem 6.2 (Classification). *Let M_0 be a closed connected smooth manifold with*

$$\dim M_0 \geq 5,$$

and put

$$\pi = \pi_1(M_0).$$

Then the assignment

$$(W; M_0, M_1) \longmapsto \tau(W, M_0)$$

induces a bijection

$$\left\{ \begin{array}{l} \text{diffeomorphism classes relative to } M_0 \\ \text{of smooth h-cobordisms over } M_0 \end{array} \right\} \cong \text{Wh}(\pi).$$

Sketch of proof. Surjectivity is Theorem 6.1. For injectivity we use the gluing formula for torsion, together with realization and the vanishing theorem [1, Chapters 2–3].

Suppose that $W : M_0 \rightsquigarrow M_1$ and $W' : M_0 \rightsquigarrow M'_1$ have the same incoming torsion x . The notation $\rightsquigarrow$ records the two ordered boundaries. If $i_0 : M_0 \rightarrow W$ and $i_1 : M_1 \rightarrow W$ are the inclusions, put

$$T_W = (i_1)_*^{-1}(i_0)_* : \text{Wh}(\pi_1 M_0) \longrightarrow \text{Wh}(\pi_1 M_1).$$

For an h-cobordism V over M_1 , additivity gives

$$\tau(W \cup_{M_1} V, M_0) = \tau(W, M_0) + T_W^{-1} \tau(V, M_1).$$

Choose $V : M_1 \rightsquigarrow N$ by realization with $\tau(V, M_1) = -T_W x$. Then $W \cup V$ has zero torsion and is a product relative to M_0 . Use its product diffeomorphism to identify N with M_0 . With this terminal identification, $T_V = T_W^{-1}$.

We can now glue V to W' along M_0 . Its torsion is

$$\tau(V \cup_{M_0} W', M_1) = -T_W x + T_V^{-1} x = 0.$$

Consequently both $W \cup V$ and $V \cup W'$ are product cobordisms, relative to their respective incoming boundaries. By associativity of gluing, after smoothing corners,

$$W \cup (V \cup W') \cong (W \cup V) \cup W'.$$

Absorbing a product collar identifies the left side with W and the right side with W' , relative to M_0 . Hence W and W' represent the same class. Notice that V is chosen with the required torsion; reversing W alone would generally introduce its dual torsion rather than its negative. $\square$

7 Boundary Torsion

The product criterion concerns the total cobordism relative to its incoming boundary. Simple homotopy equivalence of the two boundary manifolds is a different condition. The following calculation, from [1, Exercise 3.29], makes the distinction precise.

7.1 Duality

Let $j_k : M_k \rightarrow W$ be the two boundary inclusions, including any given boundary parametrizations, and put

$$\begin{aligned}\tau_k &= (j_k)_*^{-1} \tau(j_k) \in \text{Wh}(\pi_1 M_k), \\ v &= (j_0)_*^{-1} (j_1)_* : \text{Wh}(\pi_1 M_1) \longrightarrow \text{Wh}(\pi_1 M_0).\end{aligned}$$

The parametrizations are diffeomorphisms and have zero torsion for the canonical smooth simple homotopy types. Thus τ_k is the torsion of W relative to M_k , transported to the corresponding boundary group.

To include nonorientable manifolds, let $w : \pi_1(M_0) \rightarrow \{\pm 1\}$ be the orientation character. Define the anti-involution and the conjugate transpose by

$$\overline{\sum_g n_g g} = \sum_g n_g w(g) g^{-1}, \quad A^* = (\overline{a_{ji}})_{ij}.$$

These induce the involution $*$: $\text{Wh}(\pi_1 M_0) \rightarrow \text{Wh}(\pi_1 M_0)$, $*[A] = [A^*]$. For orientable manifolds, $w = 1$. With $n = \dim W$ and the odd-to-even convention, h-cobordism duality gives [1, Chapter 3][2]

$$v(\tau_1) = (-1)^{n-1} * \tau_0.$$

The exponent is the dimension of the boundary, $n - 1$.

7.2 The boundary equivalence

Choose a homotopy inverse $r_1 : W \rightarrow M_1$ of j_1 and set $h = r_1 j_0 : M_0 \rightarrow M_1$. Then $j_1 h \simeq j_0$. Homotopy invariance and the composition formula yield

$$(j_1)_* \tau(h) + \tau(j_1) = \tau(j_0), \quad \tau(h) = v^{-1}(\tau_0) - \tau_1.$$

Consequently

$$\begin{aligned}\tau(h) = 0 &\iff \tau_0 = v(\tau_1) \\ &\iff \tau_0 = (-1)^{n-1} * \tau_0 \\ &\iff \tau_0 + (-1)^n * \tau_0 = 0.\end{aligned}$$

Notice that v and $*$ have different domains: v transports from the outgoing boundary, whereas $*$ acts on the incoming Whitehead group.

Proposition 7.1. *If $\tau_0 \neq 0$ and $\tau_0 + (-1)^n * \tau_0 = 0$, then W is nontrivial relative to M_0 , but the boundary equivalence $h : M_0 \rightarrow M_1$ is simple.*

Proof. The preceding calculation gives $\tau(h) = 0$, so the Whitehead criterion makes h simple. If W were a product relative to M_0 , Theorem 5.1 would give $\tau_0 = 0$, contrary to the hypothesis. $\square$

A Algebraic Lemmas

In this appendix we record two algebraic facts used in the definition of Whitehead torsion.

Throughout this appendix, let R be an associative ring with unit and invariant basis number, as in Section 4. The complexes are bounded complexes of finitely generated free left R -modules, unless explicitly stated otherwise, and are homologically graded:

$$\cdots \longrightarrow C_{n+1} \xrightarrow{c_{n+1}} C_n \xrightarrow{c_n} C_{n-1} \longrightarrow \cdots.$$

A.1 The parity isomorphism

Let C_* be a finite contractible chain complex of finitely generated free R -modules. A chain contraction is a family of maps

$$\gamma_n : C_n \longrightarrow C_{n+1}$$

such that

$$c_{n+1}\gamma_n + \gamma_{n-1}c_n = \text{id}_{C_n}$$

for every n . In short notation,

$$c\gamma + \gamma c = \text{id}.$$

Define the odd and even parts of C_* by

$$C_{\text{odd}} = \bigoplus_i C_{2i+1}, \quad C_{\text{even}} = \bigoplus_i C_{2i}.$$

The differential c lowers degree by 1, while the contraction γ raises degree by 1. Therefore both c and γ change parity. Hence

$$c + \gamma$$

has two parity pieces:

$$u = (c + \gamma)_{\text{odd}} : C_{\text{odd}} \longrightarrow C_{\text{even}},$$

and

$$v = (c + \gamma)_{\text{even}} : C_{\text{even}} \longrightarrow C_{\text{odd}}.$$

Lemma A.1. *For every contraction γ of a bounded complex C_* , the map*

$$(c + \gamma)_{\text{odd}} : C_{\text{odd}} \longrightarrow C_{\text{ev}}$$

is an isomorphism. The contraction need not satisfy $\gamma^2 = 0$.

Proof. With u and v as above, compute

$$vu = (c + \gamma)^2|_{C_{\text{odd}}}.$$

But

$$(c + \gamma)^2 = c^2 + c\gamma + \gamma c + \gamma^2.$$

Since $c^2 = 0$ and $c\gamma + \gamma c = \text{id}$, we get

$$(c + \gamma)^2 = \text{id} + \gamma^2.$$

Thus

$$vu = \text{id}_{C_{\text{odd}}} + \gamma^2,$$

and similarly

$$uv = \text{id}_{C_{\text{even}}} + \gamma^2.$$

Now C_* is finite, and γ^2 raises degree by 2. Therefore some power of γ^2 is zero:

$$(\gamma^2)^N = 0$$

for N sufficiently large. Thus γ^2 is nilpotent. Hence

$$\text{id} + \gamma^2$$

is invertible, with inverse

$$(\text{id} + \gamma^2)^{-1} = \text{id} - \gamma^2 + (\gamma^2)^2 - \cdots + (-1)^{N-1}(\gamma^2)^{N-1}.$$

Therefore both vu and uv are isomorphisms. A left inverse of u is $(vu)^{-1}v$, and a right inverse is $v(uv)^{-1}$. They coincide, so u itself is an isomorphism:

$$(c + \gamma)_{\text{odd}} : C_{\text{odd}} \xrightarrow{\cong} C_{\text{even}}.$$

□

If the modules C_n have chosen ordered bases, then C_{odd} and C_{even} also have induced bases. Therefore this isomorphism is represented by an invertible square matrix over R . The contraction-independence theorem cited in Section 4 shows that its K_1 class does not depend on γ . After passing to $\text{Wh}(\pi)$ for $R = \mathbb{Z}\pi$, it gives Definition 4.1.

A.2 The cone criterion

Let

$$f : C_* \longrightarrow D_*$$

be a chain map between the bounded free complexes under discussion. Write d_C and d_D for their differentials. The mapping cone of f is

$$\text{cone}(f)_n = D_n \oplus C_{n-1},$$

with differential

$$\partial : \text{cone}(f)_n \longrightarrow \text{cone}(f)_{n-1}$$

given by

$$\partial(y, x) = (d_D y + f(x), -d_C x),$$

where

$$y \in D_n, \quad x \in C_{n-1}.$$

Lemma A.2. *For bounded complexes of projective modules, and in particular for the finite based free complexes used here,*

$$f : C_* \longrightarrow D_* \text{ is a chain homotopy equivalence} \iff \text{cone}(f) \text{ is contractible.}$$

Proof. Suppose first that f is a chain homotopy equivalence. It induces isomorphisms on homology. The short exact sequence of complexes

$$0 \longrightarrow D_* \longrightarrow \text{cone}(f)_* \longrightarrow C_{*-1} \longrightarrow 0$$

uses the differential $-d_C$ on the shifted quotient. Its long exact homology sequence has connecting map induced by f , up to the harmless shift sign. Therefore the cone is acyclic. It is bounded and projective, so Lemma 4.2 makes it contractible. This proves the first direction without choosing compatible homotopies for a homotopy inverse.

Conversely, assume that

$$\text{cone}(f)$$

is contractible. Then there exists a chain contraction

$$S : \text{cone}(f)_n \longrightarrow \text{cone}(f)_{n+1}$$

such that

$$\partial S + S \partial = \text{id}.$$

Write

$$S(y, x) = (u(y, x), v(y, x)),$$

where

$$u(y, x) \in D_{n+1}, \quad v(y, x) \in C_n$$

for

$$(y, x) \in D_n \oplus C_{n-1}.$$

We shall construct a chain homotopy inverse

$$g : D_* \longrightarrow C_*.$$

For $y \in D_n$, define

$$g(y) := v(y, 0).$$

First we prove that g is a chain map. Apply

$$\partial S + S\partial = \text{id}$$

to

$$(y, 0) \in D_n \oplus C_{n-1}.$$

We have

$$S(y, 0) = (u(y, 0), g(y)).$$

Therefore

$$\partial S(y, 0) = (d_D u(y, 0) + f g(y), -d_C g(y)).$$

Also,

$$\partial(y, 0) = (d_D y, 0),$$

so

$$S\partial(y, 0) = S(d_D y, 0) = (u(d_D y, 0), g(d_D y)).$$

Taking the second component of

$$\partial S(y, 0) + S\partial(y, 0) = (y, 0),$$

we obtain

$$-d_C g(y) + g(d_D y) = 0.$$

Hence

$$d_C g(y) = g(d_D y).$$

So g is a chain map.

Next we prove that

$$fg \simeq \text{id}_D.$$

Taking the first component of the same equation gives

$$d_D u(y, 0) + f g(y) + u(d_D y, 0) = y.$$

Rearranging,

$$f g(y) - y = -d_D u(y, 0) - u(d_D y, 0).$$

Define

$$H_D : D_n \longrightarrow D_{n+1}$$

by

$$H_D(y) := -u(y, 0).$$

Then

$$fg - \text{id}_D = d_D H_D + H_D d_D.$$

Thus

$$fg \simeq \text{id}_D.$$

Finally we prove that

$$gf \simeq \text{id}_C.$$

Let $x \in C_m$. Then

$$(0, x) \in \text{cone}(f)_{m+1} = D_{m+1} \oplus C_m.$$

Write

$$S(0, x) = (u(0, x), h(x)),$$

where

$$h(x) := v(0, x) \in C_{m+1}.$$

So

$$h : C_m \longrightarrow C_{m+1}.$$

Now apply

$$\partial S + S\partial = \text{id}$$

to $(0, x)$. We have

$$\partial(0, x) = (f(x), -d_C x).$$

Therefore, by linearity,

$$S\partial(0, x) = S(f(x), -d_C x) = S(f(x), 0) - S(0, d_C x).$$

Taking second components gives

$$v(f(x), -d_C x) = g(f(x)) - h(d_C x).$$

On the other hand,

$$\partial S(0, x) = \partial(u(0, x), h(x)),$$

whose second component is

$$-d_C h(x).$$

Since

$$\partial S(0, x) + S\partial(0, x) = (0, x),$$

the second component gives

$$-d_C h(x) + g(f(x)) - h(d_C x) = x.$$

Therefore

$$gf(x) - x = d_C h(x) + h(d_C x).$$

Thus

$$gf - \text{id}_C = d_C h + h d_C.$$

So

$$gf \simeq \text{id}_C.$$

Thus g is a chain homotopy inverse of f . □

For later use, the cone of the identity admits the explicit contraction

$$S(y, x) = (0, y), \quad \partial(y, x) = (d_C y + x, -d_C x),$$

since $\partial S + S\partial = \text{id}$. Its torsion is zero: the cone of a based identity is simply contractible, or equivalently reduces by elementary basis operations to identity two-term summands.

For arbitrary chain complexes, the homology long exact sequence gives:

$$\text{cone}(f) \text{ is acyclic} \iff f \text{ is a quasi-isomorphism.}$$

Here a *quasi-isomorphism* means a chain map inducing isomorphisms on all homology groups. In general acyclicity does not imply contractibility. For example, the exact complex

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

cannot be contractible, since the quotient map has no $\mathbb{Z}$ -linear section. In our bounded free setting, however, Lemma 4.2 makes acyclicity and contractibility equivalent. The definition of torsion uses a contraction of a finite based free complex, so the freeness and boundedness hypotheses remain essential.

B The Handle Model

In the proof of the s-cobordism theorem, after the h-cobordism has been put into normal form, the final identification is [1, Chapters 2–3]

$$\tau(W, M_0) = \Delta(C_*(\tilde{X}, \widetilde{\partial_0 W})) = (-1)^q[A].$$

We explain this identification in detail, with the convention of Definition 4.1. The notation $\partial_0 W = M_0$ and $\partial_1 W = M_1$ denotes the incoming and outgoing boundary, respectively.

Let

$$(W; M_0, M_1)$$

be an n -dimensional smooth h-cobordism, and put

$$\pi = \pi_1(W) \cong \pi_1(M_0).$$

Choose a handle decomposition in normal form:

$$W \cong M_0 \times [0, 1] + \sum_{i=1}^r (\phi_i^q) + \sum_{j=1}^r (\phi_j^{q+1}),$$

where

$$2 \leq q \leq n - 3.$$

Associated to this handle decomposition, the handle-to-cell construction of [1, Section 2.3] gives a finite relative CW-complex

$$(X, \partial_0 W)$$

and a homotopy equivalence of pairs

$$(f, \text{id}) : (W, \partial_0 W) \longrightarrow (X, \partial_0 W).$$

Here f is the identity on $\partial_0 W$.

The model has one relative p -cell for each p -handle. With compatible orientations, lifts, and orderings, there is a based chain isomorphism

$$\Phi : C_*(\widetilde{W}, \widetilde{\partial_0 W}) \xrightarrow{\cong} C_*(\tilde{X}, \widetilde{\partial_0 W}).$$

On the left we use the *handle* complex defined in Section 3; on the right we use the cellular complex of the relative CW model. This is not a claim that the cellular complex of an arbitrary triangulation of W is isomorphic to the handle complex. The simple homotopy property of f , proved below using the gluing theorem, is what identifies the torsion of this model with the torsion of the boundary inclusion in the manifold.

B.1 Simple homotopy type

The map

$$f : W \rightarrow X$$

is constructed inductively from the handle decomposition. We give the standard comparison argument as a proof sketch: its technical input is the gluing theorem for simple homotopy equivalences of finite CW models [1, Chapter 3][3].

Proposition B.1. *The handle-to-cell equivalence f has zero Whitehead torsion.*

Sketch of proof. At the beginning, one has

$$W_{-1} = \partial_0 W \times [0, 1], \quad X_{-1} = \partial_0 W,$$

and

$$f_{-1} : W_{-1} \rightarrow X_{-1}$$

is the projection

$$\partial_0 W \times [0, 1] \rightarrow \partial_0 W.$$

This projection has zero Whitehead torsion:

$$\tau(f_{-1}) = 0.$$

Indeed, it is the product of $\text{id}_{\partial_0 W}$ with the homotopy equivalence

$$[0, 1] \rightarrow \{*\},$$

and the product formula for Whitehead torsion gives zero.

Now suppose that

$$f_{p-1} : W_{p-1} \rightarrow X_{p-1}$$

has already been constructed and satisfies

$$\tau(f_{p-1}) = 0.$$

When one attaches the p -handles to W_{p-1} , the corresponding operation on X_{p-1} is attaching p -cells. The attaching maps commute with the projection maps up to specified homotopies. One inserts mapping cylinders to obtain compatible finite CW models before applying the gluing theorem; an arbitrary handle attaching map need not factor strictly through the projection of its attaching region.

For each p -handle, the relevant local map is the projection

$$D^p \times D^{n-p} \longrightarrow D^p.$$

On the attaching region it is

$$S^{p-1} \times D^{n-p} \longrightarrow S^{p-1}.$$

Both maps have zero Whitehead torsion, again by the product formula, since they are products with the contractible factor

$$D^{n-p} \rightarrow \{*\}.$$

Thus the following attaching diagram commutes up to the homotopy used in the construction:

$$\begin{array}{ccc} \coprod S^{p-1} \times D^{n-p} & \longrightarrow & W_{p-1} \\ \downarrow & & \downarrow f_{p-1} \\ \coprod S^{p-1} & \longrightarrow & X_{p-1} \end{array}$$

together with the corresponding maps on the attached pieces

$$\coprod D^p \times D^{n-p} \longrightarrow \coprod D^p.$$

The sum formula for these compatible models gives

$$\begin{aligned} \tau(f_p) &= (u_0)_* \tau(f_{p-1}) + \sum_H (u_H)_* \tau(D^p \times D^{n-p} \rightarrow D^p) \\ &\quad - \sum_H (v_H)_* \tau(S^{p-1} \times D^{n-p} \rightarrow S^{p-1}), \end{aligned}$$

where the sums run over the p -handles, and u_0, u_H, v_H are the structural maps of the corresponding target pieces to X_p . The induced maps put all terms in $\text{Wh}(\pi_1 X_p)$. Disconnected attaching regions are handled componentwise, and empty ones contribute zero. Each projection is simple, since the disk factor collapses to a point; equivalently its torsion is zero by the product formula. By induction the first term is also zero. Hence $\tau(f_p) = 0$.

By induction, for the final map

$$f = f_n : W \rightarrow X,$$

we get

$$\tau(f) = 0.$$

□

B.2 The relative complex

Let

$$i_0 : \partial_0 W \hookrightarrow W$$

be the inclusion, and let

$$j_0 : \partial_0 W \hookrightarrow X$$

be the inclusion of the subcomplex.

Since f is the identity on $\partial_0 W$, we have

$$f \circ i_0 = j_0.$$

By the composition formula for Whitehead torsion [1, Chapter 3],

$$\tau(j_0) = \tau(f \circ i_0) = f_* \tau(i_0) + \tau(f).$$

By Proposition B.1,

$$\tau(f) = 0.$$

Therefore, after identifying Whitehead groups via f , we get

$$\tau(i_0) = \tau(j_0).$$

Now use the relative-complex formula for the inclusion of a CW subcomplex. The underlying degreewise split exact sequence of based free $\mathbb{Z}\pi$ -chain complexes is

$$0 \longrightarrow C_*(\widetilde{\partial_0 W}) \longrightarrow C_*(\tilde{X}) \longrightarrow C_*(\tilde{X}, \widetilde{\partial_0 W}) \longrightarrow 0.$$

Since

$$j_0 : \partial_0 W \rightarrow X$$

is a homotopy equivalence, the relative complex

$$C_*(\tilde{X}, \widetilde{\partial_0 W})$$

is contractible. The additivity theorem applied to the mapping cone of the inclusion, with the cone of the identity contributing zero, gives [1, Chapter 3]

$$\tau(j_0) = \Delta(C_*(\tilde{X}, \widetilde{\partial_0 W})).$$

Combining this with the previous equality gives

$$\tau(W, M_0) = \tau(i_0) = \Delta(C_*(\tilde{X}, \widetilde{\partial_0 W})).$$

This explains the equality

$$\tau(W, M_0) = \Delta(C_*(\tilde{X}, \widetilde{\partial_0 W})).$$

The two-term parity calculation in Proposition 4.4 now gives $\tau(W, M_0) = (-1)^q[A]$. Thus the handle calculation uses the canonical simple homotopy type of the manifold, rather than only its ordinary homotopy type.

C Choices and Invariance

C.1 Inner automorphisms

Let G be a group; later we apply this to $G = \pi_1(Y, y)$ for a connected CW-complex Y . Let

$$c_a : G \longrightarrow G, \quad c_a(g) = aga^{-1}$$

be the inner automorphism determined by some fixed element $a \in G$.

Lemma C.1. *The inner automorphism c_a acts trivially on $K_1(\mathbb{Z}G)$ and on $\text{Wh}(G)$.*

Proof. This induces a ring automorphism

$$\mathbb{Z}G \longrightarrow \mathbb{Z}G, \quad g \longmapsto aga^{-1}.$$

By applying this automorphism entrywise, we get an induced map

$$\text{GL}_n(\mathbb{Z}G) \longrightarrow \text{GL}_n(\mathbb{Z}G).$$

Let

$$A = (r_{ij}) \in \text{GL}_n(\mathbb{Z}G).$$

Then

$$c_a(A) = (c_a(r_{ij})).$$

But for each entry $r_{ij} \in \mathbb{Z}G$ we have

$$c_a(r_{ij}) = ar_{ij}a^{-1}.$$

Hence, as a matrix,

$$c_a(A) = (aI_n)A(a^{-1}I_n).$$

So $c_a(A)$ is conjugate to A inside $\text{GL}_n(\mathbb{Z}G)$.

Now

$$K_1(\mathbb{Z}G) = \text{GL}(\mathbb{Z}G)_{\text{ab}} = \text{GL}(\mathbb{Z}G)/[\text{GL}(\mathbb{Z}G), \text{GL}(\mathbb{Z}G)].$$

In an abelianization, conjugate elements become equal. Therefore

$$[c_a(A)] = [A]$$

in

$$K_1(\mathbb{Z}G).$$

Thus the inner automorphism c_a induces the identity on $K_1(\mathbb{Z}G)$. Since

$$\text{Wh}(G) = K_1(\mathbb{Z}G) / \langle [\pm g] \mid g \in G \rangle,$$

it also induces the identity on $\text{Wh}(G)$.

Therefore

$$(c_a)_* = \text{id}_{\text{Wh}(G)}.$$

□

This is why basepoint-change maps on Whitehead groups are independent of the chosen path: two different paths between the same basepoints differ by an inner automorphism, and inner automorphisms act trivially on Whitehead groups.

C.2 Covers and basepoints

Let

$$f : X \longrightarrow Y$$

be a homotopy equivalence of connected finite CW-complexes. Replace it by a cellular approximation if needed; its torsion is unchanged. All coefficient identifications below use the isomorphism on fundamental groups induced by the chosen based lift.

Choose universal coverings

$$p_X : \tilde{X} \longrightarrow X, \quad p_Y : \tilde{Y} \longrightarrow Y.$$

Choose basepoints

$$\tilde{x} \in \tilde{X}, \quad \tilde{y} \in \tilde{Y}$$

such that

$$f(x) = y, \quad x = p_X(\tilde{x}), \quad y = p_Y(\tilde{y}).$$

Let

$$\pi = \pi_1(Y, y).$$

Using the isomorphism induced by f ,

$$\pi_1(X, x) \cong \pi_1(Y, y) = \pi,$$

we regard both cellular chain complexes

$$C_*(\tilde{X}), \quad C_*(\tilde{Y})$$

as based free $\mathbb{Z}\pi$ -chain complexes.

There is a unique lift

$$\tilde{f} : \tilde{X} \longrightarrow \tilde{Y}$$

such that

$$\tilde{f}(\tilde{x}) = \tilde{y}.$$

This lift induces a $\mathbb{Z}\pi$ -chain homotopy equivalence

$$C_*(\tilde{f}) : C_*(\tilde{X}) \longrightarrow C_*(\tilde{Y}).$$

The Whitehead torsion of f is defined by

$$\tau(f) := \tau(C_*(\tilde{f})) \in \text{Wh}(\pi_1(Y, y)).$$

At first this seems to depend on the choices. We now explain why it does not.

Lifts

Suppose we choose a different lift

$$\tilde{f}' : \tilde{X} \longrightarrow \tilde{Y}.$$

Then there exists a deck transformation $g \in \pi$ such that

$$\tilde{f}' = g \cdot \tilde{f}.$$

The equivariance changes at the same time. If the original lift is equivariant with respect to $f_{\#} : \pi_1(X, x) \rightarrow \pi$, then

$$\tilde{f}'(hz) = g f_{\#}(h) g^{-1} \tilde{f}'(z), \quad h \in \pi_1(X, x), \quad z \in \tilde{X}.$$

Thus its coefficient identification is $c_g \circ f_{\#}$. In particular, the deck transformation g is generally semilinear for the fixed left $\mathbb{Z}\pi$ -action:

$$C_*(g)(az) = c_g(a) C_*(g)(z), \quad a \in \mathbb{Z}\pi.$$

It is not in general an ordinary $\mathbb{Z}\pi$ -linear basis change, so one cannot apply the linear composition formula without transporting coefficients.

Transport the coefficient identification by c_g and use the lifted cellular bases ge in the target. The matrices of the new lifted map are then obtained by applying c_g entrywise to the old matrices. The cone torsion is consequently carried to $(c_g)_* \tau(f)$, which equals $\tau(f)$ by Lemma C.1. Returning to the original cellular lifts changes basis vectors only by group elements, whose classes vanish in $\text{Wh}(\pi)$. Therefore changing the lift leaves the Whitehead torsion unchanged.

Cellular bases

The cellular basis of $C_*(\tilde{X})$ or $C_*(\tilde{Y})$ is obtained by choosing, for each cell of X or Y , one lift to the universal cover and an orientation.

Changing the lift of a cell multiplies the corresponding basis vector by some

$$g \in \pi.$$

Changing its orientation multiplies the corresponding basis vector by

$$-1.$$

Thus the possible change of basis is by trivial units

$$\pm g, \quad g \in \pi.$$

But these are precisely killed in the passage from $K_1(\mathbb{Z}\pi)$ to $\text{Wh}(\pi)$:

$$\text{Wh}(\pi) = K_1(\mathbb{Z}\pi) / \langle [\pm g] \mid g \in \pi \rangle.$$

Therefore changing the cellular bases does not affect the element of $\text{Wh}(\pi)$.

Universal covers

If we choose other universal coverings

$$\tilde{X}' \longrightarrow X, \quad \tilde{Y}' \longrightarrow Y,$$

then uniqueness of universal covers gives covering isomorphisms. Choose their basepoint lifts and transport the deck-group actions so that these maps are equivariant:

$$\alpha : \tilde{X} \longrightarrow \tilde{X}', \quad \beta : \tilde{Y} \longrightarrow \tilde{Y}'.$$

The lifted maps fit into a commutative diagram up to the appropriate choice of lifts:

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{\tilde{f}} & \tilde{Y} \\ \alpha \downarrow & & \downarrow \beta \\ \tilde{X}' & \xrightarrow{\tilde{f}'} & \tilde{Y}'. \end{array}$$

With those transported coefficient identifications, the cellular chain complexes satisfy

$$C_*(\beta) \circ C_*(\tilde{f}) = C_*(\tilde{f}') \circ C_*(\alpha).$$

The chain maps $C_*(\alpha)$ and $C_*(\beta)$ are chain isomorphisms which only change cellular lifts and orientations. Hence they have zero Whitehead torsion. Using the composition formula, we get

$$\tau(C_*(\tilde{f}')) = \tau(C_*(\tilde{f})).$$

So changing the universal coverings does not change the torsion.

Basepoints

Let y' be another basepoint of Y . A path

$$w : y \longrightarrow y'$$

induces an isomorphism

$$c_w : \pi_1(Y, y) \longrightarrow \pi_1(Y, y')$$

by transporting loops along w .

Thus we get an induced isomorphism

$$(c_w)_* : \text{Wh}(\pi_1(Y, y)) \longrightarrow \text{Wh}(\pi_1(Y, y')).$$

If $v : y \rightarrow y'$ is another path, then c_w and c_v differ by an inner automorphism. Since inner automorphisms act trivially on Whitehead groups, we get

$$(c_w)_* = (c_v)_*.$$

Hence there is a canonical basepoint-change isomorphism

$$t(y, y') : \text{Wh}(\pi_1(Y, y)) \longrightarrow \text{Wh}(\pi_1(Y, y')).$$

Therefore the torsion transforms canonically under basepoint change. If we regard torsion as an element of the basepoint-independent Whitehead group

$$\text{Wh}(\Pi(Y)),$$

then changing the basepoint does not change $\tau(f)$. Here $\text{Wh}(\Pi(Y))$ is notation for the Whitehead groups of all basepoints identified by these canonical transports; $\Pi(Y)$ denotes the fundamental groupoid. It introduces no new numerical invariant.

The class $\tau(f)$ is therefore independent of covers, lifts, cellular bases, and basepoints under these canonical identifications. Homotopy invariance also makes it depend only on the homotopy class of f for the specified simple homotopy types.

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